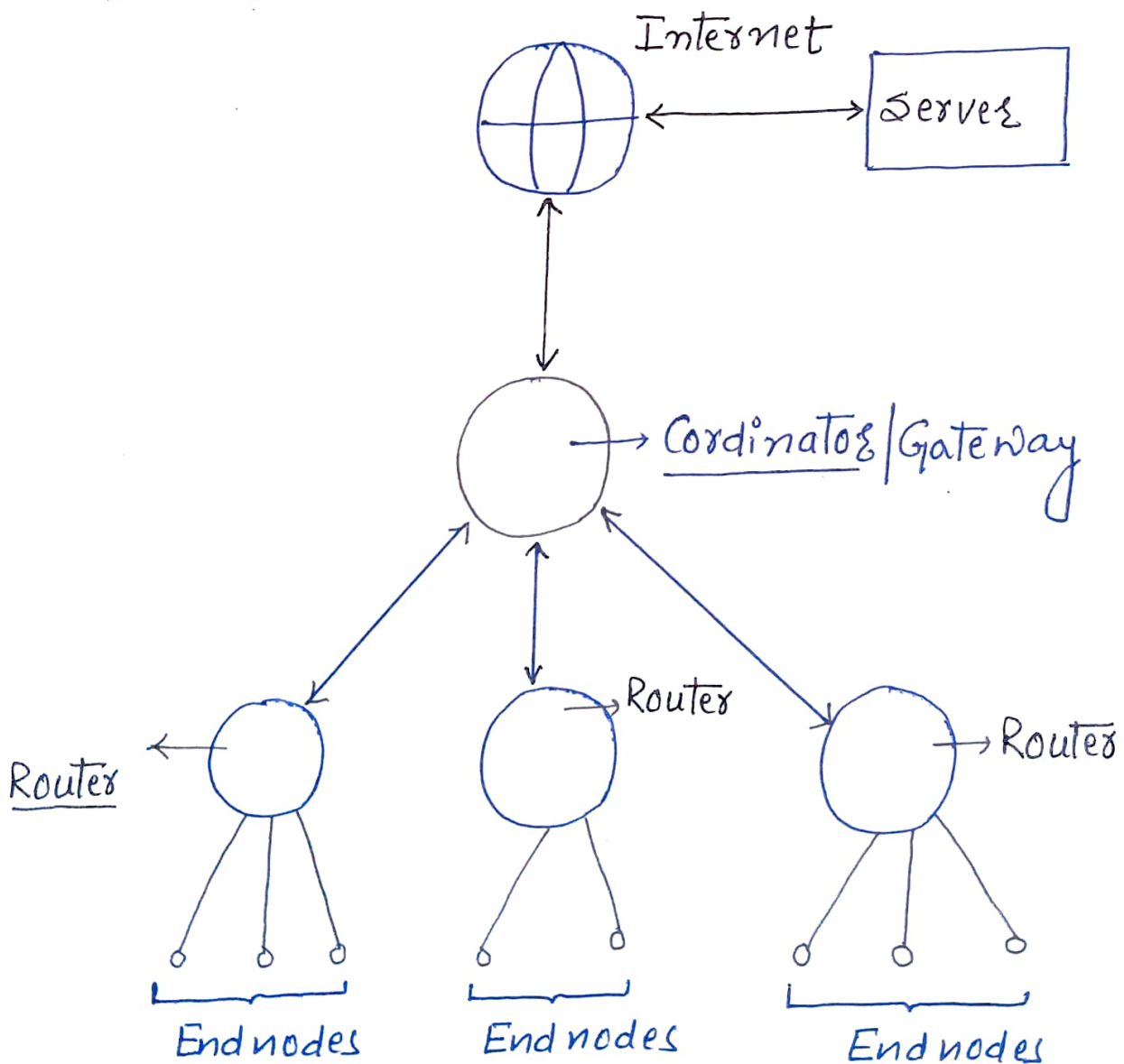


Wireless Sensor Network :- (WSN)

①

- ① WSN Comprises of distributed devices (end nodes) with sensors that are used to monitor environmental and physical conditions.
- ② WSN Consists of endnodes, routers & a Coordinator (gateway).



WSN Architecture

Working:- ① End Nodes :- End nodes use sensors to collect physical data. ②

② Now this data is transmitted to the Central system wirelessly by the end nodes. All nodes are interfaced with each other.

③ In order to transmit this physical data wirelessly, wireless protocol cards like Wi-fi, Zigbee or Bluetooth are used.

④ Once this physical data is measured, it will be routed to routers wirelessly by end nodes.

⑤ These routers are having physical data of multiple end nodes.

⑥ Now the data from routers is given to coordinator which is a gateway.

⑦ Now this gateway will pass on that data to internet where there will be server on which we will ^{be} storing all physical data.

⑧ User will be using this data after downloading wireless sensor network application in which

user will be having all the data related to all the sensors. eg:- What is the quality of air etc. ③

Characteristics of WSN:-

- ① It uses Wireless protocol cards (zigbee, Wifi, Bluetooth etc)
 - ② Range (10m-100m)
 - ③ WSNs are self organizing
 - ④ Robust (Even if some nodes get failed, system will not get failed, will auto-reconfigure itself)
 - ⑤ Uses frequency (ISM Bands - 2.4 GHz)
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Applications:- ① Soil moisture monitoring system use WSNs to monitor soil moisture at various locations.

② Indoor air quality monitoring systems use WSNs to collect data of the air quality & concentration of gases.